ECOMMERCE OPERATIONS & ANALYST
INTERN ASSIGNMENT

March 2026

DOCUMENT INFO	
Role	Ecommerce Operations & Analyst Intern
Number of Parts	3 (Q1: Inventory Excel | Q2: Ads Optimization | Q3: Operational Judgement)
Estimated Time	4–5 hours total (can be split across sessions)
Deliverables	Q1_Solution.xlsx | Q2_Solution.xlsx | Q3_Written_Responses.pdf or .docx
Submission Deadline	To be communicated separately
Data Files Provided	Q1_Inventory_Data.xlsx | Q2_AdCampaign_Data.xlsx

⚠ READ BEFORE STARTING: This assignment is designed to test real operational thinking. Some data is intentionally messy, incomplete, or contradictory — just like the real world. There is no single 'correct' answer for many questions. What matters is your reasoning, your assumptions, and whether your decisions hold up under scrutiny. Show your work.

PART 1 — INVENTORY HEALTH & REPLENISHMENT PLANNING

File provided: Q1_Inventory_Data.xlsx

Sheets: Inventory_Raw | Constraints & Context

Background

You are the Ecommerce Operations Analyst for a home décor brand selling across Amazon and Wayfair. Your inventory manager has exported a raw data file from the OMS. It is your job to clean it, analyse it, and produce a replenishment decision. The company is entering peak season in 15 days — getting stock levels right is directly tied to revenue.

Task 1.1 — Data Cleaning & Anomaly Flagging

In a new tab called Cleaned_Data, produce a deduplicated, corrected version of the raw inventory sheet. For each change you make, add a comment or a 'Change Log' column explaining what was fixed and why.

Specifically, you must identify and resolve:

- Duplicate rows — which to keep and why
- Inconsistent SKU naming across channels (same product, different SKU format)
- Missing stock values — document your assumption on how to treat them
- The negative stock SKU — what does this indicate operationally and what action does it require?
- The listing suppression case — why is it a problem and what is the operational impact?

Task 1.2 — Rate of Sale & Demand Forecast

In a new tab called RoS_Forecast, calculate the following for each valid SKU-Channel combination:

Metric	Notes
30-Day Rate of Sale (RoS)	Adjust for returns: Net = Gross - (Returns × 0.33). Returns are spread over 90 days.
Peak-Adjusted Forecast (Daily)	Apply the 40% peak uplift per Constraints sheet. Use the TikTok note for MH-LG-3301.
Days of Stock Remaining	Use Cleaned_Data stock values. Flag any SKU with < 15 days cover.
Stock Risk Category	Critical: <7 days | At Risk: 7–20 days | Healthy: 21–60 days | Overstock: >60 days

Important:

- MH-LG-3301 has a TikTok-driven sales spike. Treat the L30D figure as the new normal RoS baseline, not a weighted average of all 90 days.

- MH-MR-9901 has negative stock. Per the Constraints sheet, do NOT include it in replenishment. Document how you handle it in your RoS tab.
- Where stock is missing (MH-SC-4402 Amazon), state your assumption explicitly in a cell comment.

Task 1.3 — Replenishment Decision Table

In a new tab called Replenishment_Plan, produce a replenishment order table with the following columns:

- SKU | Channel | Risk Category | Days of Cover | Units to Order | Unit Cost | Total Cost | Priority (1–3) | Decision Notes

Constraints you must respect (from the Constraints & Context sheet):

- Total replenishment budget: $15,000
- Maximum spend per SKU: $4,000
- Amazon FBA combined stock cap: 450 units (across all FBA SKUs combined — watch your totals)
- MH-MR-9901: Do NOT replenish — investigation in progress
- MOQ must be respected for all orders — you cannot order fewer than the MOQ
- Peak season is 15 days away — any order with a lead time >15 days will NOT arrive in time. Flag these separately.

Bonus (optional, +5 marks):

- Add a Budget Utilisation summary row showing total spend, remaining budget, and FBA units committed vs cap.
- Identify which SKUs will miss peak season entirely due to lead time constraints — and suggest an alternative fulfilment approach (e.g., FBM, 3PL, partial shipment).

PART 2 — AD CAMPAIGN OPTIMIZATION & BUDGET REALLOCATION

File provided: Q2_AdCampaign_Data.xlsx

Sheets: Campaign_Raw | Budget Brief

Background

The brand ran 12 ad campaigns across Sep–Nov 2024. You have the raw campaign performance data. Going into the next month, the total ad budget is being cut from ~$37K to $6,000/month — a 40% reduction. You must decide how to allocate this budget across campaigns, document your reasoning, and communicate your decision to the client.

Task 2.1 — Metrics Calculation

In a new tab called Campaign_Analysis, calculate the following metrics for each campaign:

Metric	Interpretation Notes
CTR	Below 0.3% = poor ad creative or wrong targeting
CPC ($)	Compare across campaigns with similar product types
CVR (%)	Low CVR with decent CTR suggests listing/pricing issue
ROAS	Minimum acceptable: 1.5× (per company policy)
ACOS (%)	ACOS = 1/ROAS × 100. Target ACOS < 30%
Revenue per Click ($)	Useful for comparing efficiency across different-spend campaigns
Status Flag	Force-Pause = non-negotiable (OOS SKU or policy breach). See Budget Brief.

Note on Force-Pause:

- Any campaign linked to an out-of-stock SKU from Part 1 must be flagged as Force-Pause — running ads to a stockout wastes spend and damages ranking.
- Check your Part 1 findings. Cross-reference SKU names in campaign IDs.

Task 2.2 — Budget Reallocation Plan

In a new tab called Budget_Reallocation, produce a budget allocation table for the upcoming month:

- Total budget available: $6,000/month
- Must keep MH-BD-11500 active (client contract — non-negotiable)
- Any OOS SKU campaigns must be paused
- Run between 3 and 7 campaigns maximum

- Peak season is coming — factor in the +40% expected sales uplift

Your table must show:

- Campaign Name | Decision (Active/Paused/Force-Paused) | Allocated Budget | Rationale
- A summary row: Total Budget Used vs $6,000

AI-Proof Element: Two campaigns have conflicting signals — C006 (SD Retargeting for VS-25017) has moderate ROAS but is the only retargeting campaign, which is strategically valuable going into peak season. C010 (CU-1122 Auto) has decent volume but thin margins. You must acknowledge this tension explicitly and explain your decision — not just rank by ROAS. A mechanical sort-by-ROAS answer will score poorly.

Task 2.3 — Client Communication (Written)

Write the client-facing email (150–200 words) communicating the budget reallocation decision. You are informing the client that:

- Their total ad budget is being reduced by 40% effective next month
- Certain campaigns are being paused (list them)
- You are recommending a specific reallocation strategy with expected impact
- You are asking for sign-off before implementing

The client is not technical. Write clearly, confidently, and without jargon. Do not hide bad news — frame it with context and a plan.

PART 3 — OPERATIONAL JUDGEMENT & SCENARIO RESPONSE

No data file. Written responses only.

Submit as Q3_Written_Responses (PDF or DOCX)

This section tests your ability to think, prioritize, and communicate under real operational pressure. There are no formulas here — only judgement. Responses should be structured, specific, and practical.

⚠ Important: Generic or templated answers will score very low. We have seen many AI-generated responses to these scenarios — they are identifiable by their vagueness, their perfect bullet structure, and their complete absence of trade-off acknowledgement. Write like you are actually in the situation, with real constraints and real consequences.

Scenario 3.1 — Wrong Dimensions Live on Amazon

Time: 9:14 AM on a Monday.

Situation: You are reviewing listings as part of your morning audit. You notice that MH-BD-11500 (King Oak Bed Frame) on Amazon has incorrect product dimensions in the listing — it shows 60" x 80" (Queen size) instead of 76" x 80" (King size). The listing has been live for 3 days. In the last 24 hours alone, 47 orders have been placed.

Your task:

1. Describe your immediate actions in the next 30 minutes, in order of priority.
2. What internal stakeholders do you loop in, when, and what do you tell them?
3. How do you handle the 47 affected orders — and what do you communicate to those customers?
4. What process do you put in place to prevent this from happening again?

Word limit: 300–400 words. Be specific — 'notify the team' is not an answer.

Scenario 3.2 — Warehouse Fulfilment Failure

Time: 11:40 AM, Thursday.

Situation: You receive the following email from your warehouse partner:

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From: warehouse@partner3pl.com
Subject: URGENT — Capacity Issue, MH-CU-1122
Hi, due to an unexpected equipment failure in our pick-and-pack bay, we are unable
to fulfil the 200-unit MH-CU-1122 (Cushion Cover Set) order scheduled for dispatch
today. We expect to resume normal operations in 3–4 business days. Apologies for
the inconvenience.
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Context you know:

- MH-CU-1122 has 3 pending Wayfair orders totalling 200 units, all with promised dispatch today

- Wayfair has a supplier on-time dispatch rate requirement — missing today means a penalty and potential CastleGate programme suspension
- You have 156 units of MH-CU-1122 sitting in an Amazon FBA warehouse (not ordinarily used for Wayfair fulfilment)
- There is a second warehouse partner (Partner B) you've used before, but their last fulfilment for this SKU had quality complaints
- Your manager is in a client call until 1 PM

Your task:

5. What decisions can you make right now, without your manager? What must you wait for?
6. Draft the reply you send to the warehouse partner (3PL) — be firm, specific, and professional.
7. What options do you have for fulfilling the 200 units today or tomorrow — and what are the trade-offs of each?
8. How do you communicate with the Wayfair partner portal about the potential delay — before it becomes a breach?

Word limit: 350–450 words total across all four sub-questions.

Scenario 3.3 — Review Intelligence & Product Insight

You have been asked to review the last 90 days of customer reviews for MH-VS-25017 (Terracotta Vase) on Wayfair. Below is a representative sample:

Rating	Date	Review Text
2★	Nov 2024	Beautiful in photos but the colour is much more orange than terracotta. Returned it.
1★	Nov 2024	Arrived cracked. Packaging was clearly insufficient — just bubble wrap, no foam corners.
3★	Oct 2024	Nice shape but smaller than expected. The dimensions listed say 10in but mine measured 9.2in. Misleading.
2★	Oct 2024	Packaging damaged. The box had a hole in it and the vase had a chip on the rim.
5★	Oct 2024	Gorgeous piece, exactly as described. Will buy again.
1★	Sep 2024	Third one I've ordered — the first two arrived broken. Giving up. Packaging is the problem.
3★	Sep 2024	Good quality ceramic but the listing says 'terracotta' which I assumed was more earthy red. It's more burnt orange. Different expectations.
2★	Sep 2024	Measurement discrepancy again — listed as 10in height but feels smaller. Wish sellers would just be accurate.
4★	Sep 2024	Lovely vase. Came well packed this time. Slight colour variation from photo but I expected that.
2★	Oct 2024	Damaged on arrival. I can see why — there's basically no padding in the box for a ceramic item.

Your task:

9. Identify the distinct complaint themes (categorise them — do not list reviews one by one).

10. For each theme, assess: Is this a listing problem, a product problem, a logistics/packaging problem, or a perception/expectation problem? Explain the distinction.
11. Write the internal escalation note (100–150 words) you would send to the merchandising and supply chain teams, summarising findings and recommending specific actions.
12. One reviewer left a 4-star review mentioning 'slight colour variation from photo but I expected that.' What does this tell you that the 1- and 2-star reviews do not? How does this change (or not change) your recommendation?

This is the kind of insight translation that separates an analyst from a data entry clerk. Demonstrate it.

MARKING CRITERIA & SUBMISSION NOTES

Criterion	Marks Available	What We Are Looking For
Q1 — Data Cleaning Quality & Reasoning	20	Every change documented with business rationale, not just 'fixed'
Q1 — RoS Calculation Accuracy	15	Correct formula, correct peak adjustment, edge cases handled (spike, negative stock, missing)
Q1 — Replenishment Decisions & Trade-off Reasoning	25	Budget, MOQ, FBA cap and lead time constraints all respected. Decisions defended, not just calculated.
Q2 — Metrics Calculation Accuracy	15	All 7 metrics correct, Force-Pause correctly applied to OOS campaigns
Q2 — Budget Reallocation Judgement	15	Acknowledges SD retargeting vs ROAS tension. Does not mechanically sort by ROAS.
Q2 — Client Email Quality	10	Clear, client-appropriate language. No jargon. Specific actions. Clear ask for sign-off.
Q3 — Operational Responses (all 3 scenarios)	35	Specific actions, not generic. Trade-offs acknowledged. Real urgency and prioritisation demonstrated.
Bonus (optional)	+10	Q1 budget utilisation summary + late-arrival SKU alternatives
TOTAL	**135 (+10 bonus)**	

WHAT DIFFERENTIATES A GREAT SUBMISSION

- Decisions are made and justified — not deferred or hedged
- Edge cases are noticed without being prompted (the spike, the suppressed listing, the Force-Pause link between Q1 and Q2)
- Written communication is clear, purposeful, and audience-appropriate
- The candidate treats the data as real — not as a homework exercise
- Assumptions are documented, not hidden

A technically correct spreadsheet with no reasoning in the decision columns will score lower than a spreadsheet with minor formula errors but strong, well-documented operational thinking. We are hiring an analyst, not a calculator.

Good luck.

We look forward to seeing how you think.